

# MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 3

**1.** *Find the inverses of the following invertible matrices:*

(1) $\begin{bmatrix} 8 & 6 \\ 5 & 4 \end{bmatrix}$

(2) $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

**Solutions will be posted at the end of Week 4.**

(1) In this case, since the matrix is $2 \times 2$ there are at least two legitimate ways to do this. The first is directly by the formula for the inverse of a $2 \times 2$ matrix:

$$A^{-1} = \frac{1}{8 \cdot 4 - 6 \cdot 5} \begin{bmatrix} 4 & -6 \\ -5 & 8 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ -\frac{5}{2} & 4 \end{bmatrix}$$

The other method is to use the Gauss-Jordan algorithm (which is valid for square matrices of any size).

$$\left[\begin{array}{cc|cc} 8 & 6 & 1 & 0 \\ 5 & 4 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \to \frac{1}{8}R_1} \left[\begin{array}{cc|cc} 1 & 3/4 & 1/8 & 0 \\ 5 & 4 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \to R_2 - 5R_1} \left[\begin{array}{cc|cc} 1 & 3/4 & 1/8 & 0 \\ 0 & 1/4 & -5/8 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \to 4R_2} \left[\begin{array}{cc|cc} 1 & 3/4 & 1/8 & 0 \\ 0 & 1 & -5/2 & 4 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - \frac{3}{4}R_2} \left[\begin{array}{cc|cc} 1 & 0 & 2 & -3 \\ 0 & 1 & -5/2 & 4 \end{array}\right]$$

We conclude once again that $A^{-1} = \begin{bmatrix} 2 & -3 \\ -5/2 & 4 \end{bmatrix}$.

(2) Here we use the Gauss-Jordan algorithm:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \to R_2 - R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

We conclude that the inverse is $A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

**2.** *Is the sum of two invertible matrices always an invertible matrix? Justify your answer.*

No. Take for instance $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$. They are both invertible, since $A^{-1} = A$ and $B^{-1} = B$, and the sum $A + B$ is 0, that is a non invertible matrix.

**3.** *If a matrix $A$ is invertible, is the matrix $A^2$ invertible? Justify your answer.*

Yes. If $A$ is invertible, there is a matrix $C$ such that $AC = CA = I$. Then, if we consider $A^2$, the matrix $D = C^2$ satisfies that

$$A^2D = AACC = AIC = AC = I.$$

Similarly,

$$DA^2 = CCAA = CIA = CA = I.$$

Thus $D$ is the inverse of $A^2$, which implies that $A^2$ is invertible.

**4.** *Let $A$ be an invertible $n \times n$-matrix and $\mathbf{b}$ be a vector in $\mathbb{R}^n$. How many solution does $A\mathbf{x} = \mathbf{b}$ have? Explain why!*

Only one solution. First of all, $A\mathbf{x} = \mathbf{b}$ has *at least* one solution, $\mathbf{x} = A^{-1}\mathbf{b}$. To see that $A\mathbf{x} = \mathbf{x}$ has *at most* one solution, suppose that $\mathbf{x}_1, \mathbf{x}_2$ are solutions, i.e., $A\mathbf{x}_1 = A\mathbf{x}_2 = \mathbf{b}$. Multiplying through by $A^{-1}$ we see that $\mathbf{x}_1 = \mathbf{x}_2 = A^{-1}\mathbf{b}$, i.e., all solutions *must* be equal to $A^{-1}\mathbf{b}$ and to each other. Hence there is also *at most* one solution.